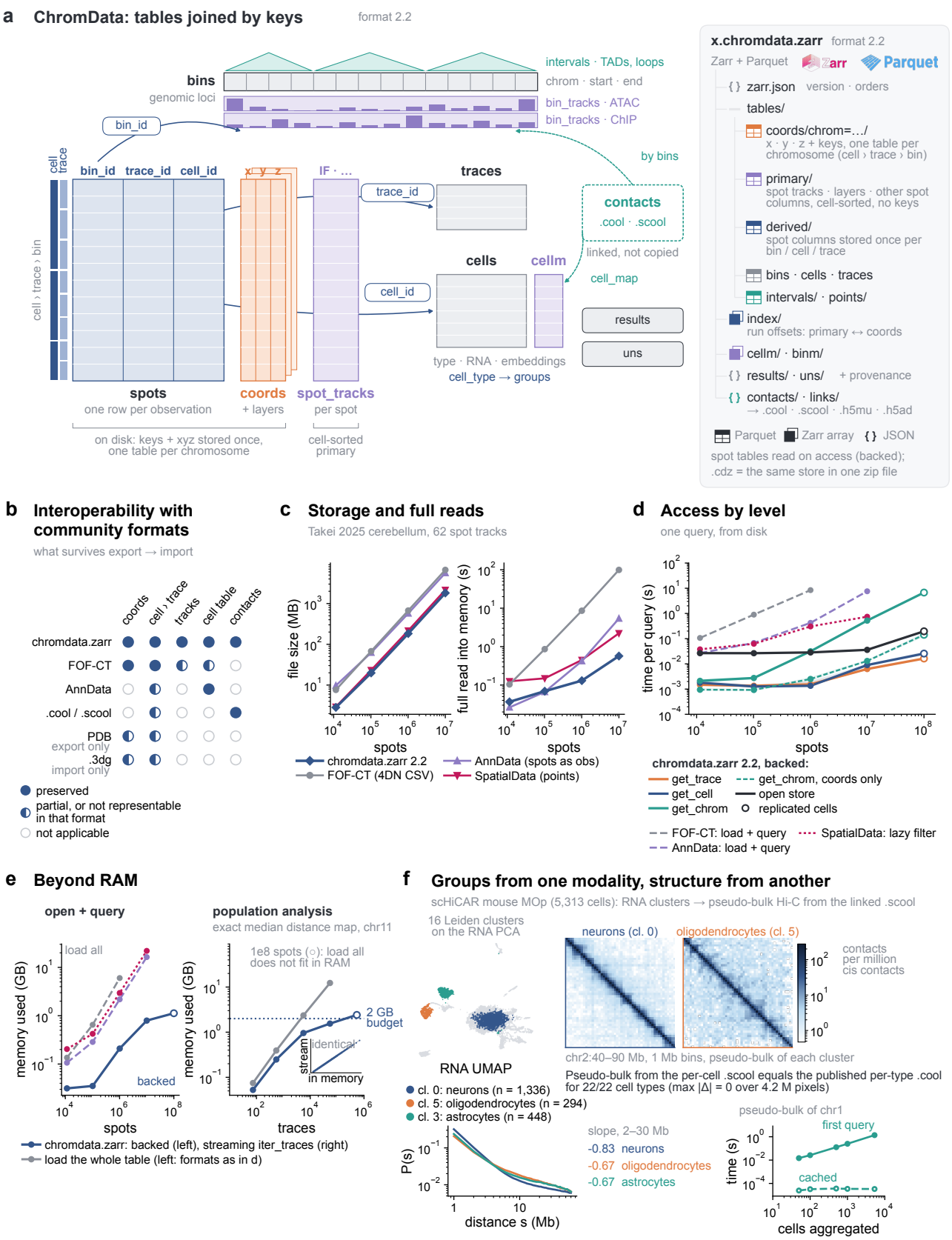

a
ChromData: tables joined by keys
format 2.2
bins
genomic loci
intervals · TADs, loops
chrom · start · end
bin_tracks · ATAC
bin_tracks · ChIP
bin_id
by bins
cell
trace
bin_id
trace_id
cell_id
x y z
IF · …
traces
trace_id
contacts
.cool · .scool
linked, not copied
cell › trace › bin
cells
cellm
cell_id
cell_map
results
uns
type · RNA · embeddings
cell_type → groups
spots
one row per observation
coords
+ layers
spot_tracks
per spot
on disk: keys + xyz stored once,
one table per chromosome
cell-sorted
primary
x.chromdata.zarr
format 2.2
Zarr + Parquet
zarr
Parquet
zarr.json
version · orders
tables/
coords/chrom=…/
x · y · z + keys, one table per
chromosome (cell › trace › bin)
primary/
spot tracks · layers · other spot
columns, cell-sorted, no keys
derived/
spot columns stored once per
bin / cell / trace
bins · cells · traces
intervals/ · points/
index/
run offsets: primary ↔ coords
cellm/ · binm/
results/ · uns/
+ provenance
contacts/ · links/
→ .cool · .scool · .h5mu · .h5ad
Parquet
Zarr array
JSON
spot tables read on access (backed);
.cdz = the same store in one zip file
b
Interoperability with
community formats
what survives export → import
coords
cell › trace
tracks
cell table
contacts
chromdata.zarr
FOF-CT
AnnData
.cool / .scool
PDB
export only
.3dg
import only
preserved
partial, or not representable
in that format
not applicable
c
Storage and full reads
Takei 2025 cerebellum, 62 spot tracks
file size (MB)
full read into memory (s)
spots
chromdata.zarr 2.2
FOF-CT (4DN CSV)
AnnData (spots as obs)
SpatialData (points)
d
Access by level
one query, from disk
time per query (s)
spots
chromdata.zarr 2.2, backed:
get_trace
get_chrom, coords only
get_cell
open store
get_chrom
replicated cells
FOF-CT: load + query
SpatialData: lazy filter
AnnData: load + query
e
Beyond RAM
open + query
load all
backed
memory used (GB)
spots
population analysis
exact median distance map, chr11
1e8 spots (○): load all
does not fit in RAM
2 GB
budget
stream
identical
in memory
traces
chromdata.zarr: backed (left), streaming iter_traces (right)
load the whole table (left: formats as in d)
f
Groups from one modality, structure from another
scHiCAR mouse MOp (5,313 cells): RNA clusters → pseudo-bulk Hi-C from the linked .scool
16 Leiden clusters
on the RNA PCA
RNA UMAP
cl. 0: neurons (n = 1,336)
cl. 5: oligodendrocytes (n = 294)
cl. 3: astrocytes (n = 448)
neurons (cl. 0)
oligodendrocytes (cl. 5)
contacts
per million
cis contacts
chr2:40–90 Mb, 1 Mb bins, pseudo-bulk of each cluster
Pseudo-bulk from the per-cell .scool equals the published per-type .cool
for 22/22 cell types (max |Δ| = 0 over 4.2 M pixels)
P(s)
distance s (Mb)
slope, 2–30 Mb
-0.83 neurons
-0.67 oligodendrocytes
-0.67 astrocytes
pseudo-bulk of chr1
first query
cached
time (s)
cells aggregated